

ECN 594: Oligopoly and Product Differentiation

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Plan

1. **Cournot competition (refresher from ECN 532)**
2. Bertrand paradox and escaping it
3. Hotelling model

From ECN 532: Oligopoly models

- You covered Cournot and Bertrand in Hector's class
- Today: quick refresher

Cournot competition: setup

- n firms producing homogeneous goods
- Firms choose **quantities** simultaneously
- Inverse demand: $P = P(Q)$ where $Q = \sum_{i=1}^n q_i$
- Constant marginal cost: c
- Firm i profit: $\pi_i = P(Q) \cdot q_i - c \cdot q_i$

Worked example: Cournot duopoly

- **Setup:** Inverse demand is $P = 100 - Q$. Two firms with $MC = 10$.
- **Steps:**
 1. Write down firm i 's profit function
 2. Find firm i 's best response function
 3. Use symmetry to find equilibrium Q and P

Worked example: Cournot duopoly (solution)

- **Step 1:** Profit: $\pi_i = (100 - q_i - q_j)q_i - 10q_i$
- **Step 2:** FOC: $\frac{d\pi_i}{dq_i} = 100 - 2q_i - q_j - 10 = 0$
- Best response: $q_i = \frac{90 - q_j}{2} = 45 - 0.5q_j$
- **Step 3:** Symmetry: $q_1 = q_2 = q^*$
- $q^* = 45 - 0.5q^* \Rightarrow 1.5q^* = 45 \Rightarrow q^* = 30$
- $Q = 60, P = 100 - 60 = 40$
- Profit per firm: $\pi = (40 - 10) \times 30 = 900$

Cournot with n symmetric firms

- **Setup:**
- n firms with constant marginal cost c
- Linear demand: $P = a - bQ$ where $Q = q_1 + q_2 + \dots + q_n$
- **Question:** What are the Cournot equilibrium quantities and price?

Cournot with n symmetric firms (solution)

- Firm i profit: $\pi_i = q_i(a - bQ) - cq_i$
- FOC: $\frac{d\pi_i}{dq_i} = a - bQ - bq_i - c = 0$
- By symmetry: $q_1 = q_2 = \dots = q_n = q^*$, so $Q = nq^*$
- Substituting: $a - bnq^* - bq^* - c = 0$
- Solving: $q^* = \frac{a-c}{b(n+1)}$, $P^* = \frac{a+nc}{n+1}$
- Profit: $\pi^* = (P^* - c)q^* = \frac{a-c}{n+1} \cdot \frac{a-c}{b(n+1)} = \frac{(a-c)^2}{b(n+1)^2}$
- **Key insight:** As $n \rightarrow \infty$, $P \rightarrow c$ and $\pi \rightarrow 0$

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2. **Bertrand paradox and escaping it**
3. Hotelling model

Bertrand competition: setup

- n firms producing **homogeneous** goods
- Firms choose **prices** simultaneously
- Consumers buy from lowest-price firm
- If tie: split demand equally
- Constant marginal cost: c

Bertrand: the paradox

- **Nash equilibrium:** $p_1 = p_2 = c$ (marginal cost pricing!)
- **Why?**
 - If $p_i > p_j > c$: firm i can undercut and capture entire market
 - Undercutting continues until $p = c$
- **The “paradox”:**
 - Only 2 firms, but competitive outcome!
 - Zero profits with just 2 competitors
 - Seems unrealistic for most markets

The Bertrand trap: real-world examples

- **Encyclopedias:**

- Encyclopedia Britannica: until the 1990s, 32-volume hardback sold for \$1,600
- Microsoft Encarta entered in the 1990s on CD for less than \$100
- By 2000: both sold for \$89.99

- **Airlines:**

- In 1992, American Airlines introduced a “value pricing” plan that cut fares
- Competitors announced even bigger cuts; American undercut further; industry followed
- Total cost of the price war: \$4 billion

Potential solutions to the Bertrand paradox

- How could we change the assumptions of the Bertrand model to get a more realistic model with positive profits (i.e. firms pricing above MC)?

Potential solutions to the Bertrand paradox

- How could we change the assumptions of the Bertrand model to get a more realistic model with positive profits (i.e. firms pricing above MC)?
- **Asymmetric (i.e. different) costs**
 - Like 'cost leadership'
- **Product differentiation/branding**
 - Undercutting the price by a small amount may no longer deliver all of total demand
- **Dynamic competition**
 - What if rivals can retaliate? E.g. you set a high price but threaten to retaliate next time if your rival undercuts you?
- **Capacity constraints**
 - What good is undercutting your rival to get total demand if you cannot actually supply all this demand due to capacity constraints?

Connection between Cournot and Bertrand

- Two-stage game:
 1. Stage 1: Firms choose capacities (quantities)
 2. Stage 2: Firms compete in prices
- **Result:** Equilibrium outcome = Cournot!
- **Intuition:**
 - Capacity choice commits firms
 - Price competition is constrained by capacity
 - Undercutting is limited by what you can produce
- Key insight: commitment matters

When does each model apply?

- **Cournot applies when:**
 - Capacity constraints matter
 - Firms commit to production before selling
 - Quantities are hard to adjust quickly
- **Bertrand applies when:**
 - Prices adjust quickly
 - No capacity constraints
 - Homogeneous products

Industry examples: which model?

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Airlines		

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Gasoline stations	Bertrand	Prices adjust daily; commodity
Online retail	Bertrand	Instant price changes; no capacity

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Static Models of Oligopoly: Hotelling Competition

- Hotelling competition relates to selling products with **horizontal differentiation**.
- 'Horizontal differentiation' relates to characteristics where there is no clear quality ordering
- Examples:
 - Geographical differentiation (one supermarket might be closer to where you live than another)
 - Product characteristics that appeal to different 'tastes' (Pepsi vs Coke)
 - Ice-cream vendors on a 1 mile beach (this was the original example given when this model was written)
- Contrast: 'Vertical differentiation' - consumers agree on which is the better product (i.e. quality differentiation)

Static Models of Oligopoly: Horizontal Differentiation On Campus



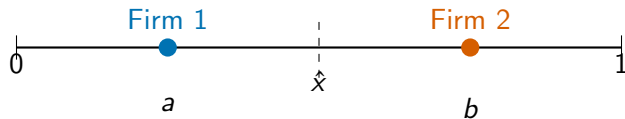
Hotelling model: setup

- Consumers uniformly distributed on $[0, 1]$ (“Main Street”)
- Two firms located at positions a and b on $[0, 1]$
- Consumer at location x has utility:

$$u_j = v - p_j - t|x - \ell_j|$$

- v : base value of product
- p_j : price of firm j
- t : transport cost per unit distance
- $|x - \ell_j|$: distance to firm j

Hotelling: graphical intuition



- Consumers to the left of $\hat{x}$ buy from Firm 1
- Consumers to the right of $\hat{x}$ buy from Firm 2
- $\hat{x}$ is the “indifferent consumer”

Solving Hotelling – Step 1: Get the payoffs

- Assume consumers always buy from one firm (high enough v)
- Consumer at x' is indifferent between the two firms if:

$$tx' + p_1 = t(1 - x') + p_2$$

- Consumers to the left of x' buy from Firm 1, to the right from Firm 2
- Solving for x' gives demand:

$$q_1 = 0.5 + \frac{p_2 - p_1}{2t}, \quad q_2 = 0.5 + \frac{p_1 - p_2}{2t}$$

- Payoffs:

$$\pi_1 = \left(0.5 + \frac{p_2 - p_1}{2t}\right) (p_1 - c_1), \quad \pi_2 = \left(0.5 + \frac{p_1 - p_2}{2t}\right) (p_2 - c_2)$$

Solving Hotelling – Step 2: Get the best responses

- Take derivative of payoffs w.r.t. own price and set to 0:

- Firm 1:

$$0.5 + \frac{p_2 - p_1}{2t} - \frac{1}{2t}(p_1 - c_1) = 0 \quad \Rightarrow \quad p_1 = \frac{1}{2}(c_1 + t + p_2)$$

- Firm 2:

$$0.5 + \frac{p_1 - p_2}{2t} - \frac{1}{2t}(p_2 - c_2) = 0 \quad \Rightarrow \quad p_2 = \frac{1}{2}(c_2 + t + p_1)$$

- Best responses are *upward sloping* (strategic complements)

Solving Hotelling – Step 3: Nash equilibrium

- Set $c_1 = c_2 = c$. By symmetry, $p_1 = p_2 = p^*$:

$$p^* = \frac{1}{2}(c + t + p^*) \Rightarrow p^* = c + t$$

- **The greater the product differentiation (t), the greater the market power**
- If $t = 0$: consumers flock to lowest price $\rightarrow$ Bertrand solution ($p = c$)
- So product differentiation ($t > 0$) solves the Bertrand paradox
- Each firm gets half the market: $q_1 = q_2 = 1/2$
- Profit: $\pi = (p^* - c) \cdot \frac{1}{2} = \frac{t}{2}$

Worked example: Hotelling with asymmetric locations

- **Question:** Two ice cream vendors on a beach of length 1 mile. Firm 1 at location 0, Firm 2 at location 0.8. Transport cost $t = 2$, marginal cost $c = 1$.
- (a) Find the indifferent consumer and each firm's demand
- (b) Find the equilibrium price for each firm
- (c) Which firm earns more profit? Explain intuitively.

Worked example: Hotelling (solution)

- **(a)** Indifferent consumer at x' : $2x' + p_1 = 2(0.8 - x') + p_2$

$$x' = 0.4 + \frac{p_2 - p_1}{4}, \quad D_1 = x', \quad D_2 = 1 - x'$$

- **(b)** Best responses (FOCs):

$$p_1 = \frac{2.6 + p_2}{2}, \quad p_2 = \frac{3.4 + p_1}{2}$$

- Solving: $3p_1 = 8.6 \Rightarrow p_1 \approx 2.87, \quad p_2 \approx 3.13$
- **(c)** $D_1 \approx 0.47, \quad D_2 \approx 0.53$

$$\pi_1 = 1.87 \times 0.47 \approx 0.87, \quad \pi_2 = 2.13 \times 0.53 \approx 1.14$$

- Firm 2 earns more: it is closer to center, so it has a larger captive market, charges a higher price, *and* has a larger market share

Hotelling with endogenous locations

- **Two-stage game:**

1. Stage 1: Firms simultaneously choose locations a and b on $[0, 1]$
2. Stage 2: Firms simultaneously choose prices

- Solve by backward induction: first solve Stage 2 for general locations

- **Stage 2 equilibrium** (for $a \leq b$):

$$p_1^* = c + \frac{t(a + b + 2)}{3}, \quad p_2^* = c + \frac{t(4 - a - b)}{3}$$

- Demands and profits:

$$D_1^* = \frac{a + b + 2}{6}, \quad \pi_1^* = \frac{t(a + b + 2)^2}{18}$$

Worked example: Should firm 1 move toward firm 2?

- **Setup:** $t = 1$, $c = 0$. Firm 2 is at $b = 1$.
- Currently, firm 1 is at $a = 0$ (endpoints).
- Firm 1 considers moving to $a = 1/4$.
- **Question:** Does firm 1 benefit from moving closer to firm 2?
- Calculate price, market share, and profit for each location.

Worked example: Location choice (solution)

- **At endpoints** ($a = 0$, $b = 1$):
 - $p_1^* = (0 + 1 + 2)/3 = 1$, $D_1^* = 3/6 = 1/2$
 - $\pi_1^* = 1 \times 1/2 = 0.50$
- **Firm 1 moves to** $a = 1/4$ ($b = 1$):
 - $p_1^* = (1/4 + 1 + 2)/3 = 13/12 \approx 1.08$, $D_1^* = (13/4)/6 = 13/24 \approx 0.54$
 - $\pi_1^* = (13/12)(13/24) = 169/288 \approx 0.59$
- **Yes!** Moving toward rival increases both price *and* market share
- Profit rises by 17%

Minimum differentiation

- Both firms have incentive to move toward center:

$$\frac{\partial \pi_1^*}{\partial a} = \frac{t(a + b + 2)}{9} > 0 \quad (\text{always!})$$

- **Hotelling (1929):** “Principle of minimum differentiation”
 - Firms cluster together to capture more customers
 - Examples: gas stations on same corner, fast food clusters
- **Problem:** With linear transport costs, the price equilibrium breaks down when firms are too close (d'Aspremont et al. 1979)
- With **quadratic** transport costs $t(x - \ell_j)^2$: result reverses to *maximum* differentiation (firms at endpoints)
- Real markets: some clustering (convenience) but not perfect overlap

Application: Strategic trade policy

- **Background:** Airplane producers: US (Boeing); EU (Airbus)
- In 2005, the US sued the EU before the World Trade Organization, accusing the EU of subsidizing Airbus
- We'd like to know: what was the impact of these subsidies on Boeing's profitability?

Strategic trade policy: setup

- **Setup:** Suppose that we have the Hotelling setup from before, where the best responses were:

$$p_1 = \frac{1}{2}(c_1 + t + p_2), \quad p_2 = \frac{1}{2}(c_2 + t + p_1)$$

- **Questions:**
- (a) What is the Nash equilibrium with asymmetric costs?
- (b) Starting from $c_1 = c_2$, what is the change in Boeing's profits (Firm 1) for a \$1 increase in subsidies for Airbus (Firm 2)?

Strategic trade policy (solution)

- **(a)** Substituting best responses into each other:

$$p_i = \frac{2}{3}c_i + \frac{1}{3}c_j + t$$

- Profit: $\pi_i = \frac{(3t + c_j - c_i)^2}{18t}$

- **(b)** A \$1 subsidy means c_2 falls by \$1. Compute $\frac{d\pi_1}{dc_2}$:

$$\frac{d\pi_1}{dc_2} = \frac{2(3t + c_2 - c_1)}{18t}$$

- At $c_1 = c_2$: $\frac{d\pi_1}{dc_2} = \frac{6t}{18t} = \frac{1}{3}$
- Since the subsidy *decreases* c_2 : Boeing's profit falls by 33 cents per \$1 of subsidy

Key Points

1. **Cournot:** Firms choose quantities; $P > MC$; as $n \rightarrow \infty$, $P \rightarrow c$
2. **Bertrand paradox:** $P = MC$ with homogeneous products, even with 2 firms
3. **Escaping the paradox:** Capacity constraints, differentiation, collusion
4. **Product differentiation** creates pricing power
5. **Hotelling:** $p^* = c + t$ (markup = transport cost)
6. Higher t (more differentiation) $\rightarrow$ higher markup
7. Indifferent consumer: $\hat{x} = \frac{1}{2} + \frac{p_2 - p_1}{2t}$
8. Firms want to cluster (minimum differentiation) but this is fragile